Page 1 - Teta Butera Nelly

$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $b = 1$ $x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$ $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

a) predictions - $\hat{y} = xm + b1$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(-1) + 3(2) \\ (4)(-1) + 10(2) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

b) errors and cost

$$e = \hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$J = \frac{1}{n} e^T e = \frac{1}{2} \begin{bmatrix} 1 & 11 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \frac{1}{2} \left((1)(1) + (11)(11) \right)$$

$$= \frac{1}{2} (1 + 121) = \frac{1}{2} (122) = 61$$

c) gradient for m: $dJ/dm = (2/n) x^T e$

$$\begin{bmatrix} dJ/dm_1 \\ dJ/dm_2 \end{bmatrix} = \frac{2}{n} X^T e$$

$$= \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = 1 \begin{bmatrix} (1)(1) + 4(11) \\ (3)(1) + 10(11) \end{bmatrix} = 1 \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

d) Gradient for b: $dJ/db = (2/n)[1\ 1]e$

$$dJ/db = \frac{2}{2}[1\ 1]\begin{bmatrix}1\\11\end{bmatrix}$$

$$= (1)\,((1)1 + 1(11))$$

$$= (1)\,(1+11) = 1(12) = 12$$

e) Update: new = old − lr × gradient (lr = 0.01)

$$m_{new} = \begin{bmatrix}m_1\\m_2\end{bmatrix} - (lr)\begin{bmatrix}dJ/dm_1\\dJ/dm_2\end{bmatrix}$$

$$= \begin{bmatrix}-1\\2\end{bmatrix} - (0.01)\begin{bmatrix}45\\113\end{bmatrix}$$

$$= \begin{bmatrix}-1\\2\end{bmatrix} - \begin{bmatrix}0.45\\1.13\end{bmatrix} = \begin{bmatrix}-1.45\\0.87\end{bmatrix}$$

$$b_{new} = b - (lr)(dJ/db) = (1) - (0.01)(12) = 1 - 0.12 = 0.88$$

f) check that the cost went down

$$\hat{y}_{new} = \begin{bmatrix}1 & 3\\4 & 10\end{bmatrix}\begin{bmatrix}-1.45\\0.85\end{bmatrix} + 0.88\begin{bmatrix}1\\1\end{bmatrix}$$

$$= \begin{bmatrix}-1.45 + 2.61\\-5.8 + 8.7\end{bmatrix} + \begin{bmatrix}0.88\\0.88\end{bmatrix} = \begin{bmatrix}2.04\\3.78\end{bmatrix}$$

$$e_{new} = \begin{bmatrix}2.04\\3.78\end{bmatrix} - \begin{bmatrix}5\\8\end{bmatrix} = \begin{bmatrix}-2.96\\-2.22\end{bmatrix}$$

$$J_{new} = \frac{1}{2}(8.7616 + 4.9284) = \frac{1}{2}(13.69) = 6.845$$

$$m = \begin{bmatrix}-1.45\\0.87\end{bmatrix},\ b = 0.88,\ J = 6.845$$

Start $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$ $b = 0.88$ $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$ ① $Y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

a) predictions $\hat{y} = Xm + b1$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + (0.88) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(-1.45) + (3)(0.87) \\ (4)(-1.45) + (10)(0.87) \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 + 2.61 \\ -5.45 + 8.7 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$= \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

b) error and cost

$$e = \hat{y} - y = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$J = \frac{1}{n} e^T e = \frac{1}{2} \begin{bmatrix} -2.96 & -2.22 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$= \frac{1}{2} \left((-2.96)(-2.96) + (-2.22)(-2.22) \right)$$

$$= \frac{1}{2} (8.7616 + 4.9284) = \frac{1}{2} (13.69 = 6.845$$

Cost before this update $J = 6.845$

c. Gradient for m $dJ/dm = 2/n X^T e$

$$\begin{bmatrix} dJ/dm_1 \\ dJ/dm_2 \end{bmatrix} = \frac{2}{n} X^T e$$

$$= \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$= (1) \begin{bmatrix} (1)(-2.96) + (4)(-2.22) \\ (3)(-2.96) + (10)(-2.22) \end{bmatrix}$$

$$= (1) = \begin{bmatrix} -2.96 + (-8.88) \\ -8.88 + (-22.2) \end{bmatrix}$$

$$= \begin{bmatrix} -11.84 \\ -31.04 \end{bmatrix}$$

(2)

d. Gradient for b; $dJ/db = \frac{2}{n} [1]e$

$$dJ/db = \frac{2}{2} [1\ 1] \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

$$= (1)[(1)(-2.96) + (1)(-2.22)]$$

$$= (1)[-2.96) + (-2.22] = (1)(-5.18) = -5.18$$

e) Update; new = old - lr x gradient (lr = 0.01)

$$m_{new} = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} - (lr) \begin{bmatrix} dJ/dm_1 \\ dJ/dm_2 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - (0.01) \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} = \begin{bmatrix} -0.1184 \\ -0.3108 \end{bmatrix} = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

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(3) $b_{new} = b - (\alpha \cdot \partial J / \partial b) = (0.88) - (0.01)(-5.18)$

$= (0.88) - (-0.05\ldots) = 0.88 + 0.518$

$= 0.9318$

(f)(1) Check that the cost went down

$\hat{y}_{new} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + (0.9318) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$

$e_{new} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$

$J_{new} = \frac{1}{2}(3.449935 + 1.9977)$

$= \frac{1}{2}(5.447635) = 2.72238$

old Cost 6.845 And New Cost 2.72238 Smaller

~~the first~~ the result update 2

$m = \begin{bmatrix} -1.3316 \\ 1.1806 \end{bmatrix}$

$b = 0.9316$

$J = 2.7238.$

$$m = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} \quad b = 0.9318 \quad X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

a) Prediction $\hat{y} = Xm + b$

$$\hat{y} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + (0.9318) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(-1.3316) + (3)(1.1808) \\ (4)(-1.3316) + (10)(1.1808) \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

b) Error and cost

$$e = \hat{y} - y = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$J = \frac{1}{n} e^T e = \frac{1}{2} \begin{bmatrix} -1.8574 & 1.4134 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= \frac{1}{2} \left((-1.8574)(-1.8574) + (1.4134)(1.4134) \right)$$

$$= \frac{1}{2} (3.449935 + 1.9977) = \frac{1}{2} (5.447635) = 2.7238$$

$$J = 2.7238$$

c, Gradient for m: $\frac{dJ}{dm} = \left(\frac{2}{n}\right) x^T e$

$$\begin{bmatrix} dJ/dm_1 \\ dJ/dm_2 \end{bmatrix} = \frac{2}{n} x^T e$$

$$= \frac{2}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= (1) \begin{bmatrix} (1)(-1.8574) + (4)(1.4134) \\ (3)(-1.8574) + (10)(1.4134) \end{bmatrix}$$

$$= (1) \begin{bmatrix} -1.8574 + 5.6536 \\ -5.5722 + 14.134 \end{bmatrix}$$

$$= \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

d, Gradient for b: $\frac{dJ}{db} = \frac{2}{n} [1\ 1]\, e$

$$\frac{dJ}{db} = \frac{2}{2} [1\ 1] = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= (1)\ ((1)(-1.8574) + (1)(1.4134))$$

$$= (1)\ (-1.8574 + 1.4134) = (1)(-0.444) = -0.444$$

e, new = old − lr × gradient (lr = 0.01)

$$m_{new} = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} - (lr) \begin{bmatrix} dJ/dm_1 \\ dJ/dm_2 \end{bmatrix}$$

$$= \begin{bmatrix} -1.0316 \\ 1.1808 \end{bmatrix} = -(0.01) \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

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$$= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - \begin{bmatrix} 0.037962 \\ 0.085618 \end{bmatrix}$$

$$= \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}$$

$$b_{new} = b - (lr)\left(\frac{d_J}{db}\right) = (0.9318) - (0.01) - (0.444)$$

$$= (0.9318) - (-0.00444) = 0.93624$$

7. check the new cost down.

$$\hat{y}_{new} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix} + (0.93624) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1.369562 + 3.285546 \\ -5.478248 + 10.95182 \end{bmatrix} + \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$

$$= \begin{bmatrix} 2.85224 \\ 6.409812 \end{bmatrix}$$

$$e_{new} = \begin{bmatrix} 2.852224 \\ 6.409812 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix}$$

$$J_{new} = \frac{1}{2}(4.612942 + 0.167946) = \frac{1}{2}(4.780888)$$

$$= 2.3904$$

old cost 2.7238 and new cost 2.3904

$$M = \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}, \quad b = 0.93624$$

$$J = 2.3904$$